

Tuning the Lattice Stage for Scale (v9)

Size-dependent block optimum · weak scheduler · 78%/67% at low time

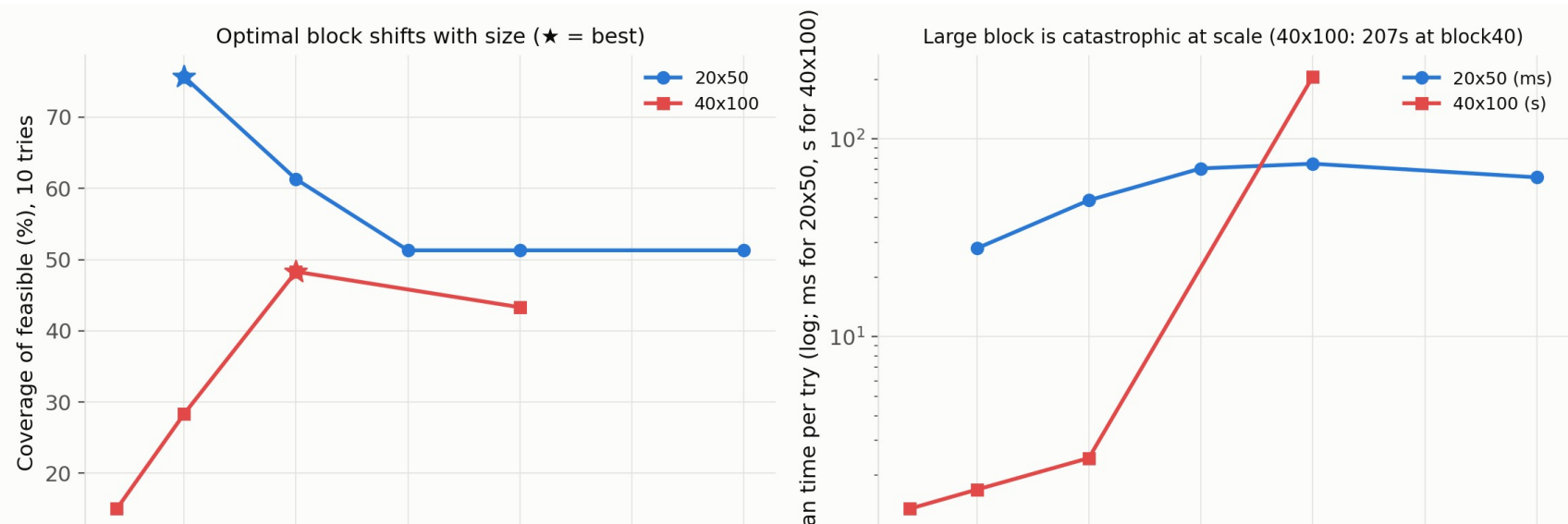
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The dominant lever: the optimal BKZ block shifts with size

20×50: coverage peaks at block 10 (75.7%), flat at 51.3% for block ≥30

40×100: unimodal, peaks at block 20 (48.3%), declines by block 40

v8 used block 40 — suboptimal at both sizes



Block 40 is catastrophic at scale: 207s per try at 40×100

BKZ cost explodes as block $\rightarrow$ lattice dimension $(n+1)$

40×100 per try: block 20 = 2.4s vs block 40 = 207s (~85× for LOWER coverage)

Scalability rule: tune block DOWN as instances grow — better coverage AND time

Tries = secondary lever (independence law, diminishing, parallelizable)

Learned scheduler rejected — signal too weak (AUC 0.54–0.58)

Best feature (first-try shortest-vector norm): AUC 0.542 (eventual), 0.576 (next-try)

Above chance but far from usable — cutting tries would discard solvable instances

Rejected on cost–benefit; fifth cross-cycle negative for learned components

Result: 78.25% at 20×50, 66.67% at 40×100, 13× faster

20×50: block 40 → 10 lifts 57.5% → 78.25%, time 4.93s → 3.78s (more, faster, fewer tries)

40×100: block 20 → 66.67% at 22.0s vs block 40 70% at 296s — 13× speedup, equal coverage

vs pre-lattice v7 (2.25%): ~30× solution-accuracy gain, now at ~3× CP-SAT time (was 37×)

Feasibility unchanged (82.5/80.0) — AHL finds solutions; refutation gap persists

Conclusion: scale comes from symbolic tuning, not learning

Scale = tune BKZ block down with size (primary) + modest parallel tries (secondary)

Both are symbolic knobs; learning does not help (fifth negative result)

Positive levers (null-space pump, lattice reduction) also scale by budget, not training

Method lesson: verify against raw artifacts — a truncated log nearly inverted this paper

Next: genuinely larger sizes ($60 \times 150+$), and the open refutation gap